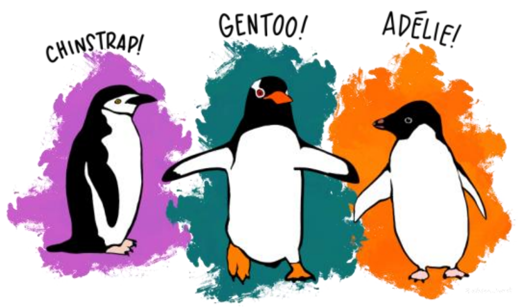
**COP2800C Module 1 Graded Programming Assignment**

## Palmer Penguins

- In this course we will design Java programs which manipulate a dataset which represents the "Palmer Penguins", a research project which tracked data about three penguin species observed on three islands in the Palmer Archipelago, Antarctica: Chinstrap, Gentoo, and Adēlie.




https://allisonhorst.github.io/palmerpenguins/

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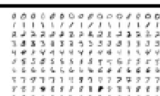
- Historically there have been some well -known and well -used (overused) datasets used for Data Science

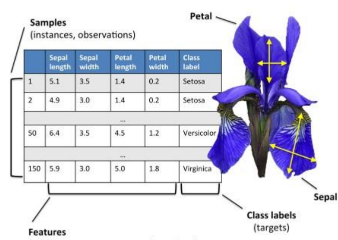
MNIST (1995)



Fashion MNIST (2017)

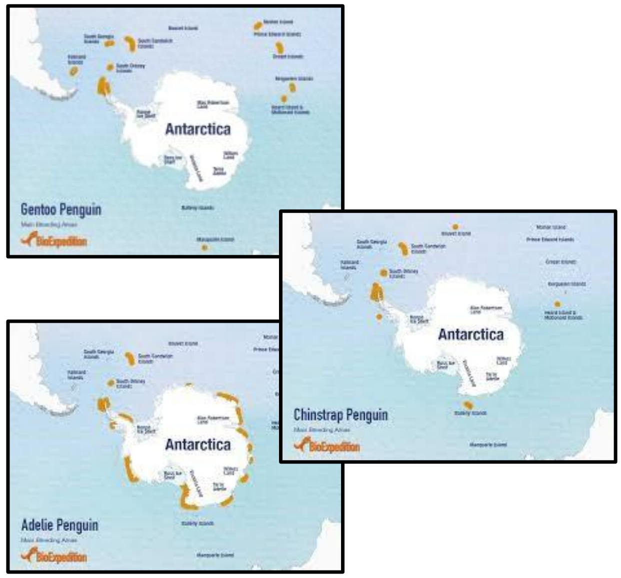
MTCARS - 32 cars from the 1973 - 74 model year. Statistics about 1974 cars: fuel consumption, design, and performance.


IRIS: Fisher's Iris data set was introduced by the British statistician, eugenicist, and biologist Ronald Fisher in a 1936 paper.



# Enter Palmer Penguins!

- Data includes size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica.
- Collected from 2007- 2009 by Dr. Kristen Gorman, Palmer Station Long Term Ecological Research Program, part of the US Long Term Ecological Research Network.
- Freely available ("No Rights Reserved").
- "The goal of Palmer Penguinsis to provide a great dataset for data exploration & visualization"




**Species**
a factor denoting penguin species (Adélie, Chinstrap and Gentoo)
**Island**
a factor denoting island in Palmer Archipelago, Antarctica (Biscoe, Dream or Torgersen)
**CulmenLength_mm**
a number denoting bill length (millimeters)
**CulmenDepth_mm**
a number denoting bill depth (millimeters)
**FlipperLength_mm**
an integer denoting flipper length (millimeters)
**BodyMass_g**
an integer denoting body mass (grams)
**Sex**
a factor denoting penguin sex (female, male)
**Year**
an integer denoting the study year (2007, 2008, or 2009)

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | StudyName | SampleNumber | Species | Region | Island | Stage | ID | ClutchCompletion | EggDate | CulmenLength-mm | CulmenDepth-mm | FlipperLength-mm | Bod Mass-g | Sex |
| 2 | PAL0708 | 1 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A1 | Yes | 11/11/2007 | 39.1 | 18.7 | 181 | 3750 | MALE |
| 3 | PAL0708 | 2 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A2 | Yes | 11/11/2007 | 39.5 | 17.4 | 186 | 3800 | FEMALE |
| 4 | PAL0708 | 3 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A1 | Yes | 11/16/2007 | 40.3 | 18 | 195 | 3250 | FEMALE |
| 5 | PAL0708 | 4 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A2 | Yes | 11/16/2007 | | | | | |
| 6 | PAL0708 | 5 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N3A1 | Yes | 11/16/2007 | 36.7 | 19.3 | 193 | 3450 | FEMALE |
| 7 | PAL0708 | 6 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N3A2 | Yes | 11/16/2007 | 39.3 | 20.6 | 190 | 3650 | MALE |
| 8 | PAL0708 | 7 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N4A1 | No | 11/15/2007 | 38.9 | 17.8 | 181 | 3625 | FEMALE |
| 9 | PAL0708 | 8 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N4A2 | No | 11/15/2007 | 39.2 | 19.6 | 195 | 4675 | MALE |
| 10 | PAL0708 | 9 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N5A1 | Yes | 11/9/2007 | 34.1 | 18.1 | 193 | 3475 | |

## Instructions

For this assignment you will write a Java program that will introduce the Palmer Penguin Species. The program will use only basic output statements (System.out.println). This is a very simple program that will give you a "quick win" and help you get familiar with the environment we will be using for the course's programming assignments. Some notes on the environment:

- **You must use Java 17 for all assignments in this course!**

- While you may be familiar with other IDEs, my demonstrations will use **jGrasp**; I will not be able to help you learn a different IDE in this course.
- Use the jGrasp IDE on the Horizon Windows 11 Academic IT system if you want to avoid installing any software.

You will submit your .java file and a screenshot (.png or .jpg only) of your program's execution by adding it to your GitHub repository as shown below. DO NOT SUBMIT .class files! (.class files are the compiled binary versions of your source code; these should <u>never</u> be submitted unless specified in the assignment instructions).

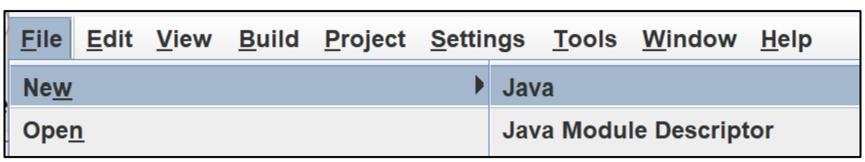
The expected output for this program is shown here; your program must duplicate this using the formatting I provide below in the walkthrough. Be sure your output is spell-checked and grammatically correct.

**Expected Output:**

```
Introducing the Palmer Penguins:
      Chinstrap!
      Gentoo!
and last but not least...
      Adelie!
There are a total of 3 penguin species in this dataset.
```

**Walkthrough**

Start with a blank Java file in the jGrasp editor (File/New/Java)




At the top of your file enter a 4-line ID header at the top of the file. You will need to do this for every assignment. Replace the placeholders in the comments (the lines starting with //) with your information. For this assignment your filename will be "PalmerPenguins.java".

**// FileName.java**
**// Your Name**
**// Submission Date**
**// 1-line summary of program's purpose (_not_ "module 1 assignment")**

Add a vertical space (a blank line), this makes your code more readable. Separate blocks of comments and code with vertical spacing (don't overdo it, however). Then declare a public Java class named "PalmerPenguins" which we will use to introduce the user to the penguins in the dataset.

```
public class PalmerPenguins {
```

Add another vertical space, then indent 4 spaces and declare your main method; this will be the program's entry point.

```
    public static void main(String[] args) {
```

Be sure to indent your code! Everything inside the class should be indented ; 4 is the preferable number of spaces for each "level" of indentation. Tabs work, but they are not always displayed consistently, spaces should be used.

Be sure to include vertical spacing (a blank line) in your code for readability. You should have a vertical space before your main method declaration.

In the main method we will display the output shown below (pay attention to the formatting). We are using **concatenation** (the '+' operator) to "glue" the string literals together in the output statements to match the formatting in the expected output.

In your main method start with a simple System.out.println statement which displays a single string literal. Include a descriptive comment above this line as shown here. Note that these lines are indented another level (a total of 8 spaces from the left margin).

```
        // output the species names with introductory text
        System.out.println("Introducing the Palmer Penguins: ");
```

In the following lines, introduce the species. Be sure to format these lines properly. Here's an example, where "\t" is used to output a tab character (using tabs in output is acceptable).

```
        System.out.println("\t" + "Chinstrap + "!");
```

Add another println statement to display the Gentoo species as shown in the expected output, then add a println to display "and last but not least..."

One more statement is needed to display Adelie with an exclamation point:

```
        System.out.println("\t" + SP_ADELIE + "!");
```

Finally, we will display the number of species. The lines (code statements and comments) in all assignments in this course must be no longer than 80 columns, so you'll need to break up the next statement into multiple lines. If you look at the status bar in jGrasp you'll see a column indicator which shows what column your cursor is currently sitting at. In the following example my cursor is at line 23, column 70 and the ASCII value of the current character is 32 (a space). "Top" represents the line number of the line currently at the top of the edit window.

Line:23 Col:70 Code:32 Top:7

Place your cursor at the end of a line to see if you need to continue the line or not. Continued lines should be indented. Break lines up at operators; don’t split a string literal if you can avoid it.

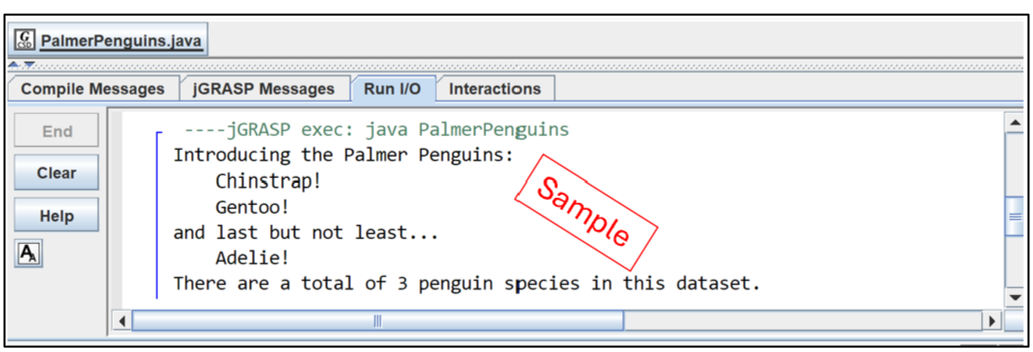
In this line we are concatenating a numeric literal between two string literals. As it would be more than 80 columns, we are splitting it at the second concatenation operator. Be sure to indent the continued line; there are various ways to do this such as using a single 4-space indentation, or using multiple indentation levels which can vary as long as the overall line is less than 80 columns. I have indented the continued line so the string literal aligns with the first argument in the println statement (another string literal).

```
System.out.println("There are a total of " + 3 +
                   " penguin species in this dataset.");
```

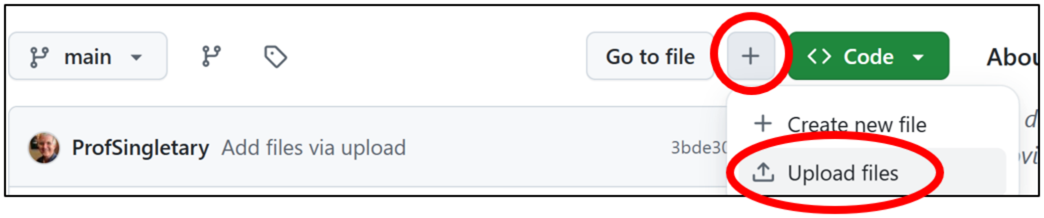
Here's a screenshot of my solution you can use as a reference. Be sure your submission uses correct indentation.

```
// PalmerPenguins.java
// D. Singletary
// 8/28/2022
// Program to introduce the Palmer Penguins

public class PalmerPenguins {

    // constants to represent the species and count
    static final String SP_CHINSTRAP = "Chinstrap";
    static final String SP_GENTOO = "Gentoo";
    static final String SP_ADELIE = "Adelie";

    static final int TOTAL_SPECIES = 3;

    public static void main(String[] args) {

        // output the species names with introductory text
        System.out.println("Introducing the Palmer Penguins: ");
        System.out.println("\t" + SP_CHINSTRAP + "!");
        System.out.println("\t" + SP_GENTOO + "!");
        System.out.println("and last but not least...");
        System.out.println("\t" + SP_ADELIE + "!");
        System.out.println("There are a total of " + TOTAL_SPECIES +
                           " penguin species in this dataset.");
    }
}
```

Screenshot the output in the "Run I/O" pane at the bottom of the jGrasp window. You may need to resize it to see everything:



Once you have completed your source code and build/executed it, use the "Upload files" option in GitHub to add your .java file and your screenshot. You can access this choice by pressing the "+" button in between the "Go to file" and "<> Code" buttons.




Again, do not submit a .class file; you should only submit your .java source code file and your screenshot.